

Credit Risk Simulation – PD, LGD & EAD

Think of a shopkeeper giving goods to a customer on credit. The big question in his mind is: “Will this person come back and pay me”?

Banks face the same question when they lend money to companies—but instead of a few hundred rupees, it's thousands of crores.

This is why **Credit Risk Modelling** is used. It breaks risk into three parts: **PD** (chance the borrower won't pay), **LGD** (how much the bank loses if that happens), and **EAD** (the total money at risk).

In this report, I've run a **Credit Risk Simulation** on four Indian listed companies to see how these risks actually look in practice.

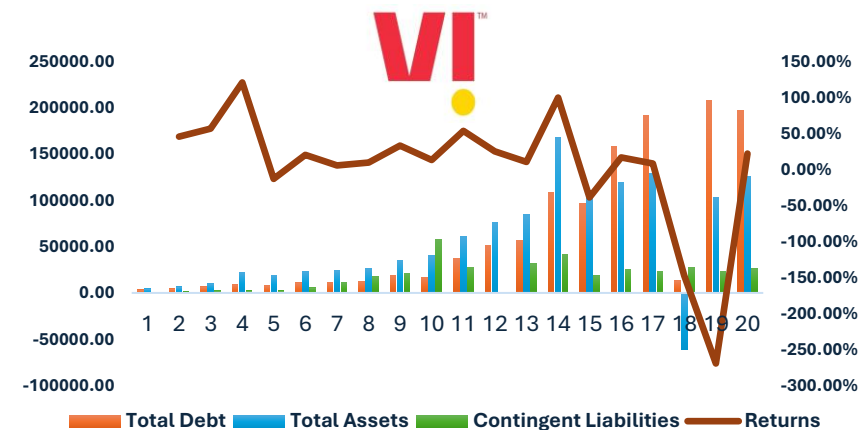
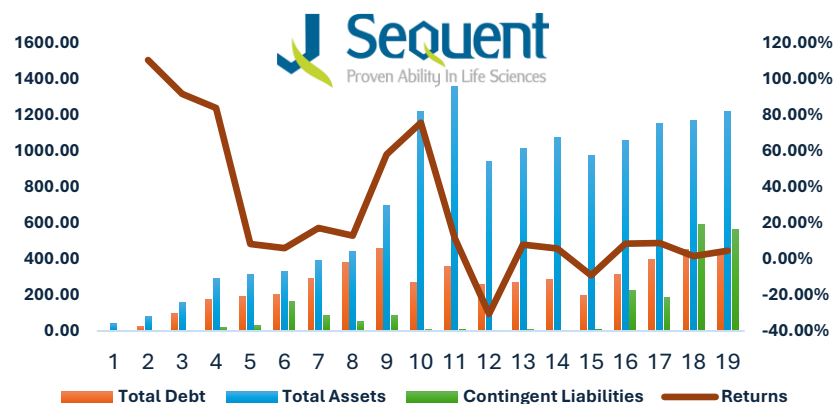
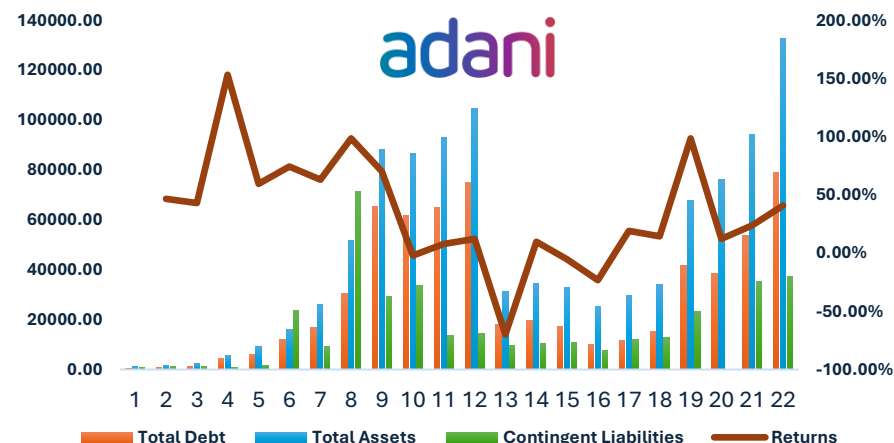
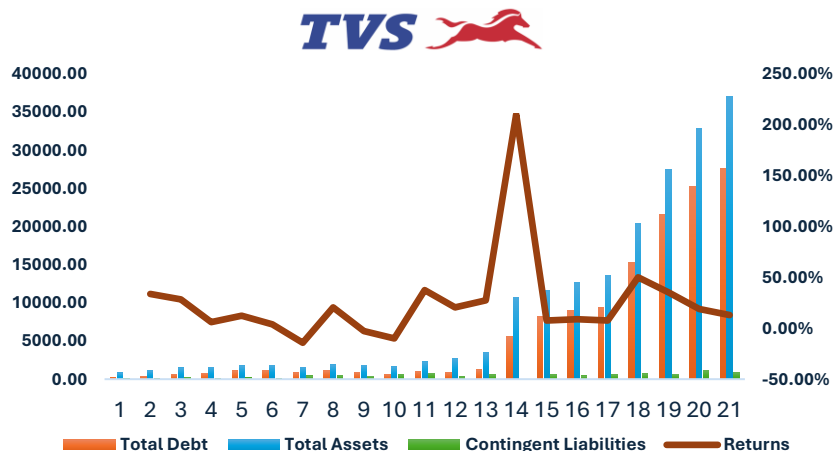
in ₹ lakh Cr	Total Assets	Total Debt	Mean	Standard Deviation
TVS Motors	37058.03	31343.92	25.73%	46.34%
Adani Enterprises	132706.9	127713.7	35.58%	48.54%
Sequent Scientific	1216.74	1106.963	26.14%	26.14%
Vodafone Idea	125976	244355	3.60%	85.49%

Shows asset strength, solid 25% gains, yet 46% volatility signals caution.

High exposures, Healthy returns (35.58%) but highly Volatile (48.54%).

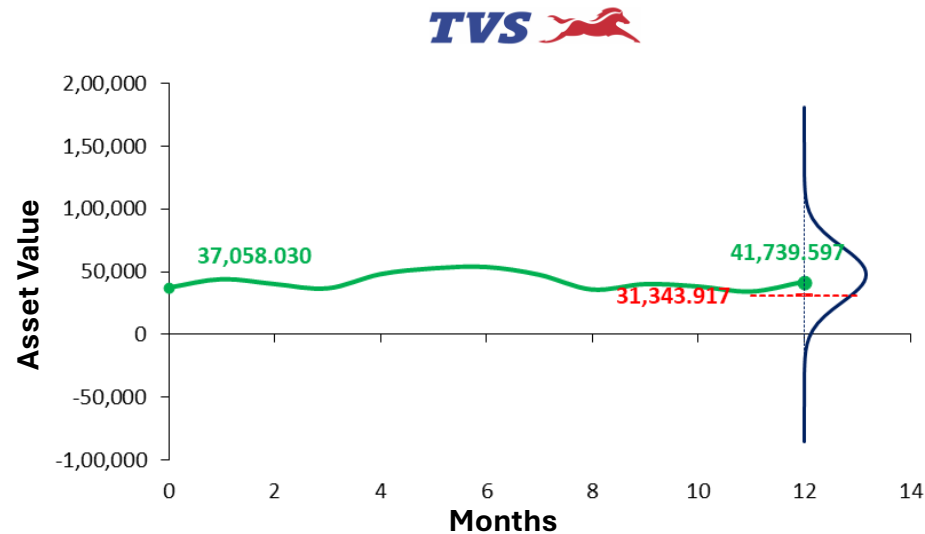
Small Company with Assets > Debt. Returns (26.14%), Volatility (26.14%)

Debt > Assets, Very low Returns (3.6%) with high Volatility (85.5%), Highly Risky.

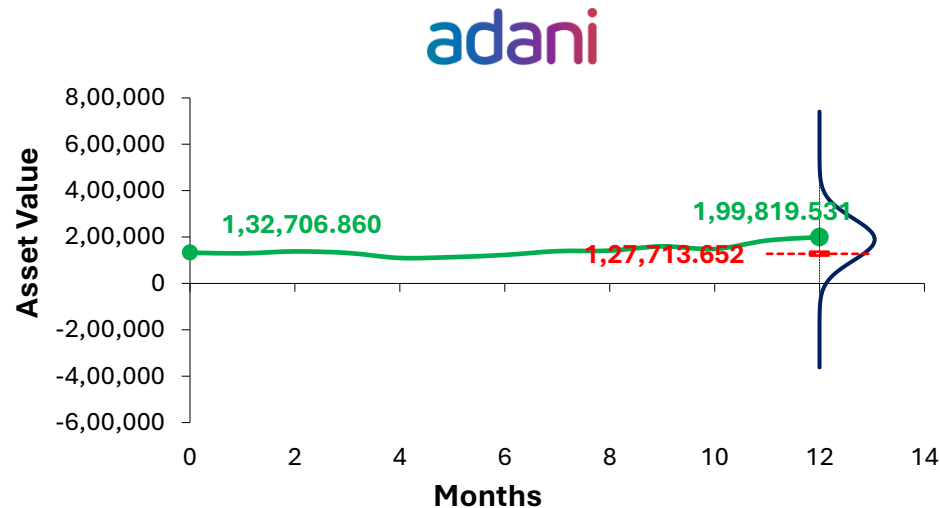


Simulation chart of four companies.

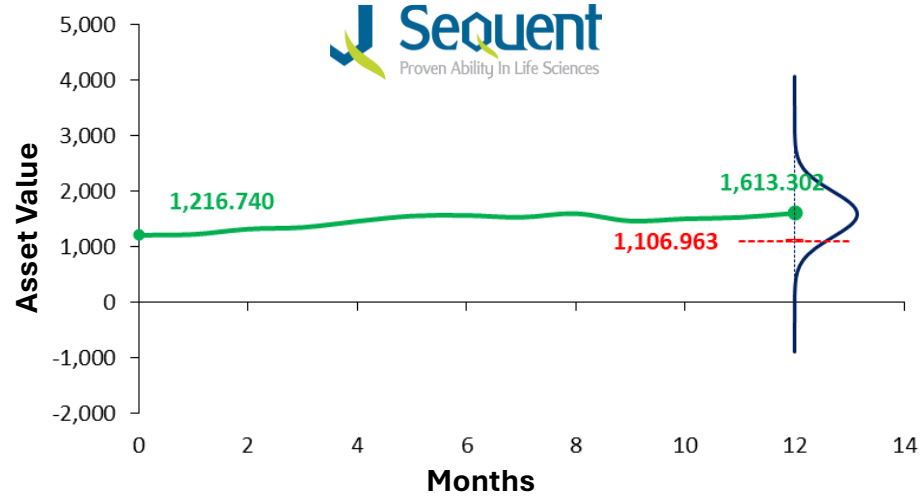
“Every company carries a different risk story — some backed by strong assets, others weighed down by heavy debt. To capture this, we’ll run **Credit Risk Simulations** for TVS Motors, Adani Enterprises, Sequent Scientific, and Vodafone Idea. By comparing how their assets move against debt over time, we’ll translate numbers into **probability of default (PD)**. In simple language, The goal is to see which companies stand steady for lenders and which ones lean toward trouble.



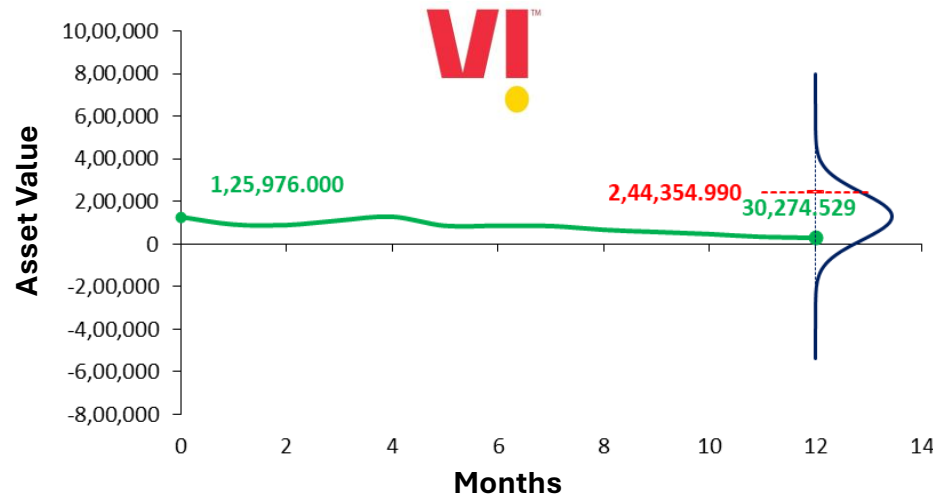
Let’s take a closer look at TVS Motors and how its assets could behave over the next 12 months. The company starts with assets around ₹37,058 Cr, while its debt stands around ₹31,343 Cr, marked by the red dashed line. Over the year, simulations (shown by the green line) suggest that assets could increase to ₹41,739 Cr in some scenarios. The blue curve on the right captures the full range of possible year-end asset values — while significant outcomes keep assets above debt, a small portion dips below, which is why TVS has a 20.1% probability of default. In simple terms, TVS Motors is not in immediate danger, but its asset fluctuations mean there’s a real chance of default. Compared to Vodafone Idea, it’s safer, yet the variability keeps it in a moderate-risk category — generally stable, but with notable downside risk in certain scenarios.



“Adani Enterprises starts the year with assets around ₹1,32,706 Cr against a debt of ₹1,27,713 Cr. The simulations show that while the company’s assets generally stay above debt, there’s a fair chance they could dip close to or below it in some scenarios. The blue curve illustrates the spread of possible year-end outcomes — most results suggest moderate stress, but high exposure combined with a slightly higher PD of 19.7% highlights elevated risk. In short, Adani is relatively stable most of the time, but the magnitude of its debt means that when things go wrong, losses could be significant, placing it in a higher-risk category than TVS Motors.



Sequent Scientific's position is comparatively safe. The company begins with assets of roughly ₹1,216 Cr and debt of ₹ 1,106 that is much lower in proportion. The simulations show that assets stay above debt in nearly all scenarios, with a tight range of outcomes reflected in the blue curve. The probability of default (PD) is 17.6%, indicating that the chance of the company failing to meet obligations is low. Even in less favorable simulations, the asset buffer is sufficient to cover debt comfortably. For lenders, Sequent Scientific represents a low-risk profile — stable and predictable, with minimal exposure to extreme losses. It's the safest among the four companies in this portfolio.



Vodafone Idea presents a stark contrast. With assets of around ₹1,25,976 Cr versus enormous debt of ₹2,44,354 Cr, simulations indicate that assets fall below debt in the vast majority of scenarios. The blue curve stretches widely, but most of the outcomes lie beneath the red debt line, signaling serious vulnerability. This is reflected in the extremely high probability of default (PD) of 85.2%. In practical terms, this means Vodafone Idea is highly likely to default, making it a very high-risk company for any lender. The magnitude of debt combined with asset volatility creates a scenario where even small unfavorable changes can lead to significant losses, placing Vodafone Idea in the top tier of portfolio risk. Most scenarios suggest distress, and only rare outcomes keep the company above debt.

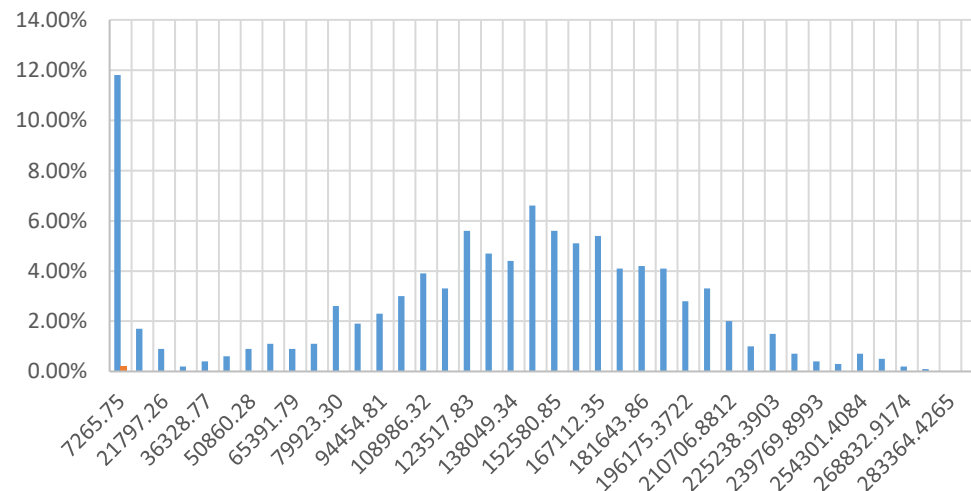
Credit loss Distribution.

After analyzing the default risk of each company individually, we now take a step back to look at the bigger picture — the **Credit Loss Distribution** of our total portfolio, which includes TVS Motors, Adani Enterprises, Sequent Scientific, and Vodafone Idea. This distribution doesn't just show potential losses; it assigns a probability to each scenario, allowing us to understand how often certain losses might occur and how extreme events could impact the portfolio. By visualizing the range of outcomes — from frequent, manageable losses to rare but severe shocks — we can clearly see the overall risk profile.

	PD	Alpha	Beta	EAD
TVS Motors	17.9%	2.94	7.06	31343.92
Adani Enterprises	19.70%	2.60	7.40	127713.65
Sequent Scientific	17.60%	2.06	7.94	1106.96
Vodafone Idea	85.20%	5.84	4.16	244354.99

Sequent Scientific	Low PD (17.6%) and very small exposure (1106.96) - Low Credit Risk
TVS Motors	Moderate PD (17.9%) with sizeable exposure (31343.92) - Moderate Credit Risk
Adani Enterprises	Slightly higher PD (19.7%) with large exposure (127713.65) - Elevated Credit Risk
Vodafone Idea	Extremely high PD (85.2%) with massive exposure (244354.99) - High Credit Risk

Credit Loss Distribution



Imagine a bank lending to these four companies - TVS Motors, Adani Enterprises, Sequent Scientific, and Vodafone Idea. How much could it lose, and how often? That's exactly what this **Credit Loss Distribution** shows. Smaller losses, around ₹7,625 Cr, are relatively common, reflecting situations where perhaps only one company struggles. Most of the outcomes sit in the middle range, between ₹1.2 lakh Cr and ₹1.8 lakh Cr, which is the 'expected zone' when one or two companies default. Then there's the tail on the far right — rare but severe losses exceeding ₹2.3 lakh Cr — representing extreme scenarios where multiple large defaults happen together. This chart gives us a clear picture of both expected and worst-case losses. Keep in mind, these numbers come from a simulation, so they're illustrative — actual results could vary."

Key Learnings

- **Probability of Default (PD):** PD provides a clear measure of how often a company might fail, but its impact must be considered alongside exposure.
- **Exposure at Default (EAD):** Large exposures magnify risk — even a moderate PD can translate into heavy losses if the exposure is significant.
- **Loss Given Default (LGD):** The severity of loss when a default occurs is critical — higher LGD means less recovery for lenders and greater overall impact.
- **Asset vs. Debt Dynamics:** A company's default risk depends on how its assets move relative to debt — strong assets can still pose risk if volatility is high.
- **Portfolio Perspective:** Beyond individual companies, the **Credit Loss Distribution** of the total portfolio reveals expected losses and extreme downside scenarios.
- **Tail Risk Matters:** Rare but severe events, such as multiple defaults, can dominate portfolio losses, making stress-testing essential.
- **Practical Takeaway:** Banks must weigh both typical outcomes and worst-case scenarios — simulations turn uncertainty into actionable insights for lending decisions.

Conclusion

Balance sheets give us snapshots, but simulations reveal the full story of risk. By testing asset values against debt for TVS Motors, Adani Enterprises, Sequent Scientific, and Vodafone Idea, we see how **Probability of Default (PD)**, **Loss Given Default (LGD)**, and **Exposure at Default (EAD)** play out in real money terms.

Vodafone Idea emerges as the riskiest — debt far outweighs assets, pushing PD above 85%. **Adani**, with its huge scale, looks safer most of the time, but even small default chances create massive potential losses. **TVS Motors** appears solid, yet volatility keeps its PD near 20%. **Sequent** is smaller and more balanced, meaning its risk is real but the impact far lighter.

The real power of **credit risk modelling** lies in clarity — it turns vague uncertainty into measurable patterns. This means lenders aren't just guessing; they can prepare for both the expected outcomes and the rare but costly surprises.